# School mapping in ESRI imagery

Andrija Gorup, Dominik Jambrović, Marin Kačan, Siniša Šegvić

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# Recap from last meeting

# Model fine-tuning

- All Anditi schools + equal number of OSM Vietnam non-schools
- Static Gaussian non-school sampling
- 20 epochs
- Train results
  - **F1: 99.55%**
  - P: 99.80%
  - R: 99.30%

# Dense inference

- Use fine-tuned model on **overlapping satellite images** (tiles) from chosen districts
- Perform non-maximum suppression (**NMS**) to reduce the number of duplicate detections
- Measure **number of detected Anditi schools (train data)**
- Deliver **list of potential schools** sorted by prediction confidence

# Dense inference results (after NMS)

- Check performance on **train data (Anditi schools)** – sanity check before evaluation
- Found 900/946 (**95.14%**) Anditi schools
- Recall: 95.14%
- 46 not detected schools (**4.86%**)
- List of potential schools: **9573 images**

# Problems

- No dataset for evaluating the final fine-tuned model
- Only measured train recall, not precision and **F1 score**

# New contributions

# New validation datasets

- We have hand-collected two subsets of school locations:
  - Exhaustive dataset
  - Wide dataset
- Can be used for evaluating the final fine-tuned model

# Exhaustive (narrow) dataset

- Locations from 4 subdistricts (wards) from **Thủ Đức District**:
  - Linh Xuan Ward
  - Binh Chieu Ward
  - Tam Binh Ward
  - Hiep Binh Phuoc Ward

- **110 schools** (exhaustive search), **1163 overlapping tiles total** (after filtering)

- Can be used for calculating precision, recall and **F1 score**

# Wide dataset

- Locations from 5 districts:
  - Lê Chân District
  - Hoài Đức District
  - Cà Mau
  - Bà Rịa
  - Sóc Trăng
- **93 schools** (not exhaustive search!), **20 981 overlapping tiles total** (after filtering)
- Can be used for calculating recall

# Threshold validation

- We want to determine the ideal classification threshold
- Get dense inference predictions for exhaustive dataset
- Measure F1 score on predictions for **variable thresholds (0-1)**
- Pick threshold with the **highest F1 score**
- Get recall, precision and F1 score for chosen threshold on **exhaustive dataset**

# Threshold validation

- Procedure is applied to multiple models with different fine-tune configurations (fine-tune on **Anditi dataset** with non-schools sampled from **OSM Vietnam**)

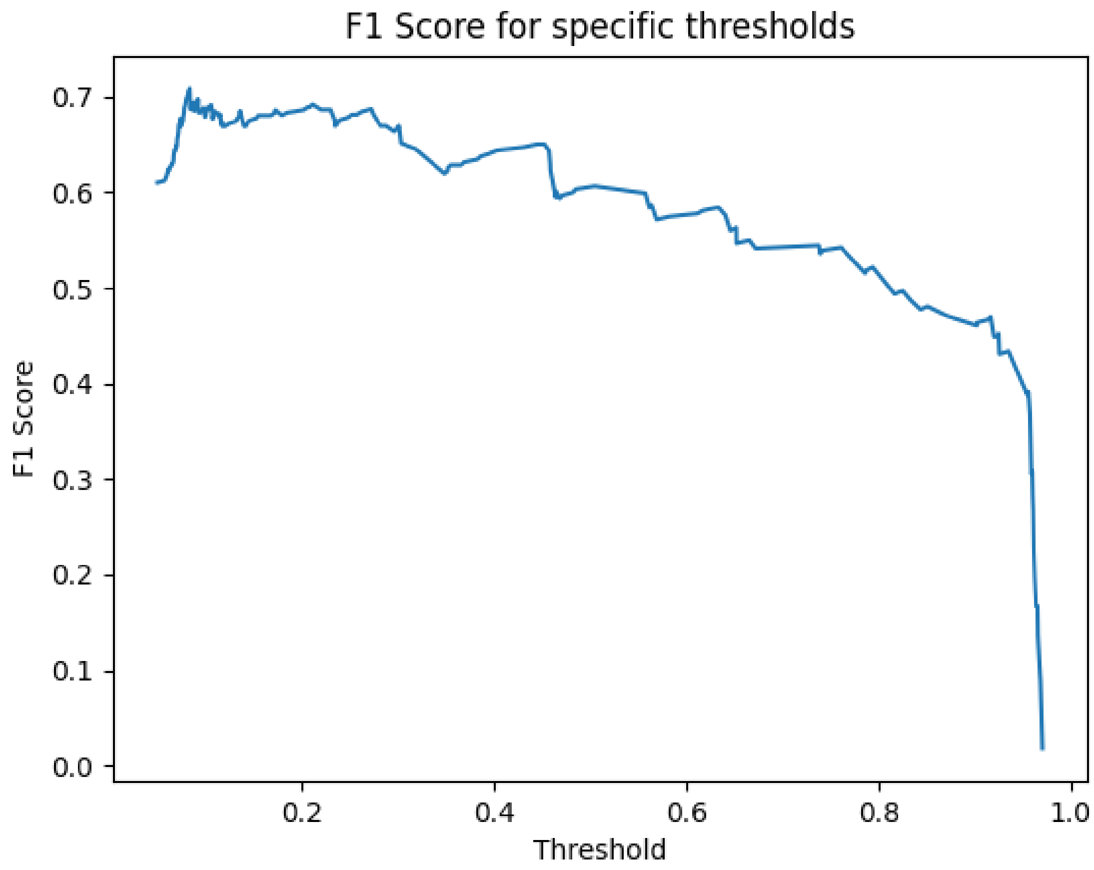
- Different number of **fine-tune epochs**:
  - 1 epoch
  - 20 epochs

- **Dynamic Gaussian** non-school sampling with different school : non-school sampling ratios:
  - 1:1
  - 1:10

# Threshold validation, results

| School : non-school ratio | Fine-tune epochs | Best threshold (%) | Best F1 score (%) | Best Recall (%) | Best precision (%) |
|---|---|---|---|---|---|
| 1:1 | 20 | 3.61 | 61.85 | **97.27** | 45.34 |
| 1:10 | 20 | 4.04 | 63.70 | 84.55 | 51.10 |
| 1:1 | 1 | 8.35 | **70.90** | 96.36 | 56.08 |
| 1:10 | 1 | 6.33 | 69.14 | 76.36 | **63.16** |

# Threshold validation, 1:1 ratio, 1 epoch



# Conclusion & Future work

- F1 score for fine-tuned model (1:1 ratio, 1 epoch):
  - Train: 92.55%
  - **Exhaustive: 70.90%**
- Fine-tuning for 20 epochs may overfit the nation-wide model to Anditi dataset
- Explore and test options for reducing overfitting